

T3 - TEARCI

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N16

Relay Based H Bridge

Transistor Driver Design (Forced $\beta \approx 10$) – Arduino Relay Interface

Each relay coil is controlled by an NPN BJT configured as a low-side switch. The device is intentionally operated only in cut-off and saturation, as required by the design specifications. A forced current gain of approximately $\beta \approx 10$ is adopted to guarantee reliable saturation under worst-case conditions.

A flyback diode is connected in parallel with each relay coil (cathode to +V_COIL and anode to the transistor collector) to clamp the inductive voltage spike generated when the transistor turns off.

Relay Coil Current

For a standard Arduino starter-kit relay such as the Songle SRD-05VDC-SL-C, the coil is rated at 5 V and has an approximate resistance of 70 Ω . Under nominal supply conditions, this results in a steady-state coil current of about 71–72 mA.

Collector Current

Since the transistor operates as a low-side switch, the collector current is essentially equal to the relay coil current when the relay is energized.

Base Drive Design (Forced $\beta = 10$)

To ensure the transistor operates in hard saturation, the base current is dimensioned according to a forced current gain of 10.

With a coil current of approximately 72 mA, the minimum required base current is about 7.2 mA.

The Arduino digital output drives the transistor base through a resistor. Considering a conservative saturation base-emitter voltage of approximately 0.8 V and a conservative minimum high-level output voltage of about 4.1 V (at 20 mA output current), the calculated base resistor value is approximately 458 Ω .

A standard resistor value of $470\ \Omega$ is therefore selected. This provides roughly 7 mA of base current under worst-case conditions and slightly higher current under typical operating conditions, ensuring robust and reliable saturation.

Power Dissipation

When the transistor is saturated, the collector-emitter voltage is typically between 0.1 V and 0.2 V with adequate base drive. Consequently, the power dissipation in the transistor remains low and well within safe operating limits.

The resulting base current also remains within the recommended per-pin current capability of an Arduino Uno output.